ML PROBLEM-FRAMING MEMO

Customer Churn Prediction System

Internship: Artificial Intelligence & Machine Learning

Organization: RabTech Academy

Project: Responsible Customer Churn Prediction

1. Business problem

Customer churn occurs when a customer stops using a company's product or service. Churn can reduce recurring revenue and increase the effort needed to acquire replacement customers.

This project explores whether customer tenure, support activity, monthly spending, login recency, and subscription plan can help identify patterns associated with churn.

2. Decision and prediction target

Target label: churned (1 = churned, 0 = not churned). The unit of observation is one customer record.

Intended decision: whether a customer may be considered for a human-reviewed retention follow-up. The model is a decision-support prototype, not an automated decision-maker.

The supplied dataset does not specify a formal prediction horizon or the exact time at which each feature was recorded. These must be established before real-world use.

3. Available input features

tenure_months: length of the customer relationship, in months.

support_tickets: number of support tickets.

monthly_spend_inr: monthly spending in INR.

last_login_days: number of days since the customer's last login.

plan_type: subscription category.

customer_id is an identifier and is excluded from predictive modelling.

4. Why consider machine learning?

A classification model can learn associations between the available customer attributes and the observed churn label. Such analysis may help prioritize human review for retention activities.

Machine learning is not automatically necessary. For a small dataset or simple business process, manual review or transparent rule-based analysis may be more appropriate.

5. Proposed modelling approach

Use Logistic Regression as an interpretable baseline classification algorithm.

Standardize numerical variables and one-hot encode the categorical plan_type variable. Exclude customer_id from model inputs.

Compare against a majority-class DummyClassifier baseline.

Use Leave-One-Out Cross-Validation for this educational dataset of 12 records. Each iteration trains on 11 records and tests on one. This provides a demonstration of evaluation, not a reliable estimate of future performance.

6. Evaluation plan

Accuracy: overall proportion of correct predictions.

Precision: proportion of predicted churn cases that are actually labelled churned.

Recall: proportion of actual churn cases identified by the model.

F1-score: harmonic mean of precision and recall.

Confusion matrix: counts of true positives, true negatives, false positives, and false negatives.

Review errors and compare the model with the majority-class baseline. Calibration and meaningful fairness comparisons require substantially more data.

7. Error costs and human review

False positive: a customer is flagged as likely to churn but does not churn. This may lead to unnecessary contact, discounts, or campaign costs.

False negative: a customer churns but is not flagged. This may cause a missed retention opportunity and potential revenue loss.

The business must determine the relative costs before choosing an operating threshold. A human reviewer should check any proposed retention action. Uncertain cases should be referred for manual review rather than treated as definite predictions.

8. Responsible use and safeguards

Do not automatically cancel accounts, deny services, or change prices based on model output.

Confirm data provenance, permission, and appropriate consent before use beyond the educational task.

Protect customer information and exclude unnecessary identifiers.

Check for data leakage and ensure input features are available at the prediction time.

Review performance, data quality, and potential differences in errors across groups when sufficient data exists.

9. Monitoring and rollback

Monitor data quality, prediction errors, and changes in customer behaviour.

Reassess the model when the data distribution or business process changes.

If predictions become unreliable, show unexpected bias, or depend on invalid data, suspend model-assisted decisions and return to manual or rule-based review until the issue is investigated and validated.

10. Limitations and conclusion

The supplied dataset contains 12 records and 7 columns, including the target. It is too small to support reliable claims about generalization, fairness, or production readiness.

This notebook and memo demonstrate a responsible baseline workflow. A larger, representative dataset, documented provenance and permissions, a defined prediction horizon, and stronger validation are required before deployment.